

Digital Asset Cryptography

Part 3 - Bitcoin: Protocol Proposals & Activation Politics

Compiled by the Spaced Research Ledger

Last updated: 2026-08-31

Bitcoin has no quantum-resistant way to hold coins today, and no activation scheduled to create one. What it has is a set of draft proposals at different levels of ambition. Add to that a live demonstration of how hard activating anything has become, and one workaround that needs no permission at all.

The proposals form a sequence. BIP-360 would create an address type with no quantum-exposed spending path. BIP-361 would eventually sunset the old signatures entirely. Both defer the hardest technical choice, which post-quantum signature scheme to adopt. Meanwhile the one contested soft fork actually in flight during 2026, unrelated to quantum, is failing to activate, which says something about the road any of this must travel.

Section 1 covers BIP-360. Section 2 covers BIP-361 and its phased sunset. Section 3 covers the fight over freezing dormant coins. Section 4 covers the signature schemes, Section 5 the activation politics, and Section 6 the no-fork workaround that reached mainnet in August.

1. BIP-360: An Address Type Without a Key Path

The nearest-term proposal removes the quantum-vulnerable spending path from Bitcoin addresses. It deliberately does not add post-quantum signatures.

Recent Developments

The proposal keeps changing after its merge. The merged text carries version 0.12.1, dated 24 July 2026, meaning it has been revised in the repository for over five months since the February merge. Merged should not be read as frozen.[1]

Current State

BIP-360's current title in the bitcoin/bips master copy is Pay-to-Merkle-Root, P2MR for short. Its status is Draft, and its authors are Hunter Beast, Ethan Heilman and Isabel Foxen Duke.[1] The output type has been renamed twice, from P2QRH to P2TSH to P2MR. The middle rename came in a clean-sheet rewrite announced in December 2025, which also moved post-quantum signatures out to a future proposal and added Python and Rust test vectors.[2]

The design commits solely to the Merkle root of a script tree rather than to an internal public key, which removes Taproot's quantum-vulnerable key-path spend entirely. The draft specifies SegWit version 2, producing addresses that begin with bc1z.[1] It supports multisig, timelocks, conditional payments, inheritance schemes and advanced custody. The cost is slightly larger transactions, since every spend carries more witness data than a Taproot key-path spend.[3]

The proposal explicitly does not introduce post-quantum signatures. Its own text defers them, saying protection against more sophisticated quantum attacks may need post-quantum signatures in the future, and it mandates no scheme.[1]

2. BIP-361: The Legacy Sunset

The more radical proposal would eventually make old signatures unspendable, on a timetable measured in years. It is a draft, not a plan the network has accepted.

Current State

BIP-361, titled Post Quantum Migration and Legacy Signature Sunset, sits in the bitcoin/bips repository as a consensus soft fork with status Draft. Its authors are Jameson Lopp, Christian Papathanasiou, Ian Smith, Joe Ross, Steve Vaile and Pierre-Luc Dallaire-Demers.[4] Its requirements point at a post-quantum signature proposal that has not been written yet.[4]

The sunset runs in phases. Phase A forbids sending funds to quantum-vulnerable address types, permitting only transfers from legacy to post-quantum addresses. It activates about 160,000 blocks, roughly three years, after BIP-361 itself does.[4] Phase B tightens signature validation at a predetermined block height roughly two years after Phase A, encumbering legacy signatures and requiring wallets to have upgraded by then.[4]

After Phase B, holders of old coins could spend only through rescue protocols built on asymmetric knowledge, such as hardened key derivation. The intent is that a genuine owner can prove ownership where an attacker holding only the public key cannot.[4]

Unresolved Conflicts

The proposal's own record disagrees about its date. The repository preamble says it was assigned on 11 February 2026.[4] Press coverage dates the public submission to 14 April 2026, when Lopp and five co-researchers submitted the draft.[5] The gap may be the difference between assigning a number and publishing the draft, but no source on record confirms that reading, so both dates stand.

3. The Freeze-or-Burn Debate

Any sunset raises the question of coins whose owners never migrate, including Satoshi's. The disagreement is about governance, not engineering.

Current State

Jameson Lopp's position is that freezing dormant coins beats letting a future attacker recover and sell them, and he calls an attacker sweeping dormant coins theft from everyone.[5] Michael Saylor argues quantum computing will not break Bitcoin but harden it. In his words the network upgrades, active coins migrate, lost coins stay frozen, security goes up and supply comes down.[6] Charles Hoskinson argues Bitcoin's quantum fix amounts to a hard fork and cannot save Satoshi-era coins. He is a rival ecosystem's principal commenting on Bitcoin, not a neutral assessor.[7]

The objection to all of these positions is governance-shaped rather than technical. Critics dispute who decides which coins are lost rather than merely dormant, and argue that any freeze is inconsistent with self-custody and a fixed supply.[6]

4. Choosing a Signature Scheme

Whatever Bitcoin eventually activates will need a post-quantum signature scheme, and every candidate costs block space. No choice has been made.

Recent Developments

A new proposal for SHRINCS, a hash-based post-quantum signature scheme optimised for Bitcoin, was published in late August 2026. Its signatures start at 548 bytes, plus a 48-byte public key, with security resting on the SHA-256 hash function.[8]

Current State

The choice is a three-way trade, and BIP-360's deferral means nobody has to make it yet.[9] ML-DSA has been standardised since August 2024 and is well studied, but takes the most block space. FN-DSA-512 would cut the on-chain footprint by roughly three times, but its standard, FIPS 206, is not yet published. SLH-DSA rests on hash-function security rather than newer mathematical assumptions, making it the conservative backstop at a still larger signature size.[9]

5. Activation Politics

Bitcoin changes by consensus among miners, developers and users, and nothing quantum-related has entered that process yet. The one contested change that has entered it is failing.

Recent Developments

BIP-110, a proposed temporary soft fork limiting certain data storage, opened its mandatory signaling window at block 961,632 around 9 August 2026, running through block 963,647. It has nothing to do with quantum risk, but it is the current working example of how a contested soft fork actually fares.[10] As of 30 July it had reached about 2% signaling against a 55% early lock-in threshold, with no major mining pool committed. The expected outcome was a small breakaway chain rather than a network-wide change.[10]

Current State

The covenant proposals sometimes linked to quantum migration run on their own timelines. OP_CAT reached complete specification status in March 2026 but has no proposed mainnet activation parameters. OP_CTV has a published activation client specifying a 90% miner threshold and a minimum activation height around May 2027.[10] Neither the covenants nor any quantum-resistant spending path has activation consensus, and their schedules are not aligned to each other. The migration itself is characterised as a multi-year

programme, the largest coordinated key rotation Bitcoin has ever attempted, and one that has not properly started.[10]

6. The No-Fork Workaround

One team decided not to wait for consensus. The result works today, costs about two hundred dollars per transaction, and proves a point rather than solving the problem.

Recent Developments

The first QSB spend was mined on Bitcoin mainnet in block 964,199, reported on 27 August 2026. Producing it cost between \$150 and \$200 in computation and took hours.[11] The transaction was nonstandard and could not propagate through normal node relay, so it had to be submitted directly through MARA Pool's Slipstream service. StarkWare characterises the technique as a last-resort measure for individual transactions, not a network-wide upgrade.[11]

Current State

QSB, short for Quantum Safe Bitcoin, was published on 9 April 2026 by Avihu Levy of StarkWare. It embeds a hash-based one-time signature scheme in Bitcoin Script, working under existing consensus rules with no protocol change, no miner signaling and no activation timeline.[12] The cost is the point of the design rather than a footnote. It requires substantial off-chain GPU computation, with estimated costs of roughly \$75 to \$200 per transaction.[13]

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from three to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

Sources

1. [github.com — bip 0360.mediawiki](https://github.com/bitcoin/bips/blob/master/bip-0360.mediawiki) - <https://github.com/bitcoin/bips/blob/master/bip-0360.mediawiki> (as of 2026-07-24)
2. [groups.google.com — 3JWQwnMUetg](https://groups.google.com/g/bitcoinddev/c/3JWQwnMUetg) - <https://groups.google.com/g/bitcoinddev/c/3JWQwnMUetg> (as of 2025-12-19)
3. [cointelegraph.com — bitcoin s quantum upgrade path what bip 360 changes and what it does not](https://cointelegraph.com/features/bitcoin-s-quantum-upgrade-path-what-bip-360-changes-and-what-it-does-not) - <https://cointelegraph.com/features/bitcoin-s-quantum-upgrade-path-what-bip-360-changes-and-what-it-does-not> (undated)
4. [github.com — bip 0361.mediawiki](https://github.com/bitcoin/bips/blob/master/bip-0361.mediawiki) - <https://github.com/bitcoin/bips/blob/master/bip-0361.mediawiki> (undated)

5. theblock.co — bitcoin researchers propose phased sunset of legacy signatures to address quantum risks - <https://www.theblock.co/post/397556/bitcoin-researchers-propose-phased-sunset-of-legacy-signatures-to-address-quantum-risks> (as of 2026-04-14)
6. bitcoinist.com — saylor lost bitcoin freeze quantum risk - <https://bitcoinist.com/saylor-lost-bitcoin-freeze-quantum-risk/> (as of 2025-12-17)
7. coindesk.com — cardano s hoskinson says bitcoin s quantum fix is a hard fork that can t save satoshi s coins - <https://www.coindesk.com/tech/2026/04/16/cardano-s-hoskinson-says-bitcoin-s-quantum-fix-is-a-hard-fork-that-can-t-save-satoshi-s-coins> (as of 2026-04-16)
8. SHRINCS BIP Published: Quantum-Secure Bitcoin Comes With A Catch - <https://cointelegraph.com/magazine/shrinCS-bip-is-published-quantum-secure-bitcoin-comes-with-a-catch> (as of 2026-08-27)
9. postquantum.com — fixing bitcoin pqc migration - <https://postquantum.com/quantum-threat-crypto/fixing-bitcoin-pqc-migration/> (undated)
10. aminagroup.com — bitcoin fork august 2026 bip 110 ecash covenants and the quantum clock - <https://aminagroup.com/research/bitcoin-fork-august-2026-bip-110-ecash-covenants-and-the-quantum-clock/> (as of 2026-08-09)
11. cointelegraph.com — starkware quantum resistant bitcoin transaction mainnet - <https://cointelegraph.com/news/starkware-quantum-resistant-bitcoin-transaction-mainnet> (as of 2026-08-27)
12. theblock.co — starkware quantum safe bitcoin - <https://www.theblock.co/post/396987/starkware-quantum-safe-bitcoin> (as of 2026-04-09)
13. coindesk.com — quantum safe bitcoin now possible without a soft fork but costs usd200 a pop - <https://www.coindesk.com/markets/2026/04/10/quantum-safe-bitcoin-now-possible-without-a-soft-fork-but-costs-usd200-a-pop> (as of 2026-04-10)